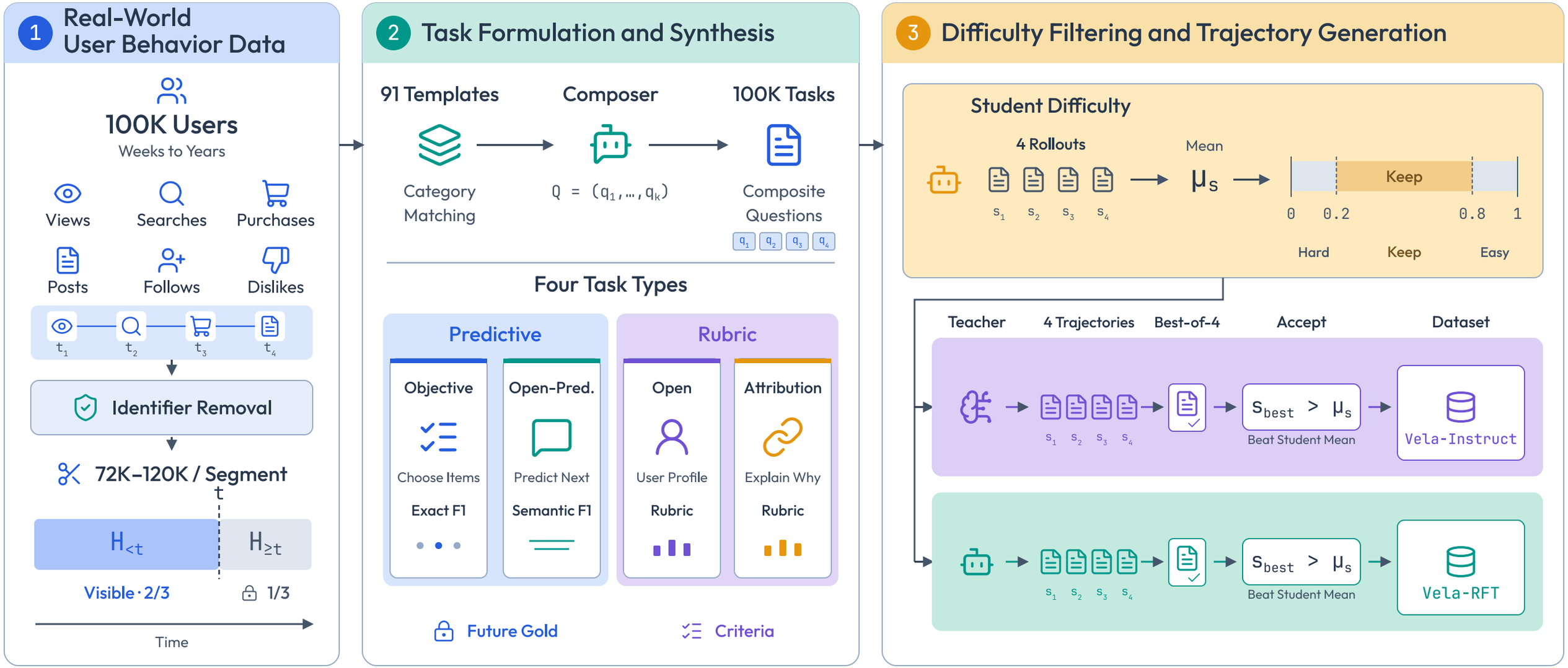

1
Real-World
User Behavior Data
100K Users
Weeks to Years
Views
Searches
Purchases
Posts
Follows
Dislikes
t1
t2
t3
t4
Identifier Removal
72K–120K / Segment
t
H<t
H≥t
Visible · 2/3
1/3
Time
2
Task Formulation and Synthesis
91 Templates
Composer
100K Tasks
Category
Matching
Q = (q1, …, qk)
Composite
Questions
q1
q2
q3
q4
Four Task Types
Predictive
Objective
Choose Items
Exact F1
Open-Pred.
Predict Next
Semantic F1
Rubric
Open
User Profile
Rubric
Attribution
Explain Why
Rubric
Future Gold
Criteria
3
Difficulty Filtering and Trajectory Generation
Student Difficulty
4 Rollouts
Mean
μs
s1
s2
s3
s4
Keep
0
0.2
0.8
1
Hard
Keep
Easy
Teacher
4 Trajectories
Best-of-4
Accept
Dataset
sbest > μs
Beat Student Mean
Vela-Instruct
Vela-RFT